Cache

# Development of a Disaster-Resilient Infrastructure Preparedness Chatbot for Flood-Prone Communities

A Retrieval-Augmented Generation (RAG) chatbot grounding flood-preparedness guidance in official Philippine government sources.

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## WHY THE SELECTED SDG IS RELEVANT?

# SDG 9: Industry, Innovation and Infrastructure

Floods repeatedly damage roads, buildings, drainage systems, and bridges in Philippine communities. Resilience here is not only physical — it also depends on communities having timely access to clear, actionable preparedness information.

A chatbot addresses the information side of resilience: closing the gap between what official agencies publish and what residents can actually find and understand before heavy rainfall hits. Research on Filipino flash-flood preparedness further shows that knowledge of flood risk, perceived severity and vulnerability, and past flood experience all shape whether households actually take preparatory action — reinforcing that access to clear information is itself a preparedness intervention.

### Infrastructure at risk from flooding

- Roads & bridges
- Residential & community buildings
- Drainage systems
- Electrical installations

## THE PROPOSED METHODOLOGY

# How Retrieval-Augmented Generation Works

RAG combines information retrieval with language generation. Before producing a response, the system retrieves relevant passages from an external knowledge source, then the language model composes an answer grounded in that retrieved material — rather than relying only on what it already knows internally.

### User Query

A resident asks a question in English, Tagalog, or Taglish.

### Retrieval

System searches the curated DOST-PAGASA / OCD-NDRRMC knowledge base.

### Grounded Context

Relevant passages are pulled and passed to the model as supporting context.

### Generated Answer

Llama 3.1 synthesizes a clear, actionable response grounded in those passages.

## WHY RAG FITS THIS PROBLEM?

- Knowledge-intensive & domain-specific: answers must reflect exact government guidelines, not general knowledge the model already holds.
- Reduces hallucination risk: responses are grounded in retrieved, verifiable passages rather than generated from memory alone.
- Updatable without retraining: the curated knowledge base can be expanded or corrected as new advisories are published.

## WHAT PROBLEM THE GROUP INTENDS TO ADDRESS?

# Narrow Problem Statement

Residents in flood-prone communities struggle to find clear, easy-to-understand guidance on protecting basic infrastructure before floods — information that exists, but is scattered across dense government and disaster-preparedness documents. The table below narrows this from a broad domain down to a concrete chatbot focus.

| | |
|---|---|
| • **Broad domain** | Infrastructure resilience and disaster preparedness. |
| • **SDG-related issue** | SDG 9 — building reliable, sustainable, resilient infrastructure that supports well-being and development. |
| • **Target users** | Residents and community members in Philippine flood-prone areas who need comprehensible risk-reduction guidance. |
| • **Narrow problem** | Lack of accessible, easy-to-understand information on drainage maintenance, infrastructure preparedness, flood-risk reduction, and pre-/during-rainfall actions. |
| • **Proposed chatbot focus** | A RAG chatbot using credible government, disaster, weather, and infrastructure sources to answer preparedness, protection, drainage, and risk-minimization queries. |

**What is the degree of success for a chatbot that uses RAG to retrieve accurate answers on topics related to flood-preparedness and infrastructure protection from a curated set of documents?**

## Focused

Evaluated only on flood-preparedness and infrastructure-protecti on questions — not general disaster topics.

## Measurable

Assessed via correctness, relevance, and groundedness of retrieved documents.

## Methodologically fit

RAG grounds answers in a defined knowledge base rather than relying on model memory alone.

WHERE THE INITIAL DATA COULD COME FROM?

# Initial Knowledge-Base Data Sources

## DOST-PAGASA

### DOCUMENT TYPE

Government webpages, flood forecasts, flood bulletins, advisories, and learning materials.

### RELEVANCE TO THE CHATBOT

Supplies flood forecasting, rainfall conditions, warning levels, and precautionary actions to answer risk and warning-related queries.

### AVAILABILITY

Highly accessible — published publicly on PAGASA's official website; materials can be selected and downloaded directly.

## OCD / NDRRMC

### DOCUMENT TYPE

Government plans, disaster preparedness guidelines, reports, advisories, and risk-reduction publications.

### RELEVANCE TO THE CHATBOT

Supplies flood forecasting, rainfall conditions, warning levels, and precautionary actions to answer risk and warning-related queries.

### AVAILABILITY

Accessible — published through OCD/NDRRMC's official platforms.

## HOW THE INITIAL DATA WILL BE USED

# Initial Evaluation Data Plan

Benchmark questions will be drawn from Barangay DRRM community orientations, public flood-preparedness FAQs, and official DOST-PAGASA / OCD-NDRRMC advisories — then paired with source passages to build ground-truth answers.

- **Factual / informational**

  e.g., standard heights for elevating electrical installations.

- **Procedural**

  e.g., neighborhood drainage-cleaning steps before heavy rainfall.

- **Scenario-based**

  Retrofitting advice weighed against community vulnerability and perceived flood severity.

- **Tagalog & Taglish phrasing**

  Authentic community-language inputs, tested alongside English.

- **Out-of-scope (boundary tests)**

  e.g., live forecasts or engineering certifications — checks fallback to official agencies.

WHY THE PROPOSED LANGUAGE MODEL IS A REASONABLE CHOICE?

# Language Model: Llama 3.1 8B Instant

## llama-3.1-8b-instant

*Deployed via the Groq LPU API platform*

**8B** *parameters*

**128k** *context window*

### Instruction-following precision

Optimized to stay aligned with strict system prompts that constrain outputs to retrieved context — critical for factual disaster guidance.

### Ultra-low latency, low cost

Groq's LPU inference removes local GPU requirements and keeps operational cost low for an academic prototype.

### Multilingual capability

Supports effective processing of Tagalog and Taglish community inputs alongside English.

### Open-weights flexibility

Enables zero-shot prompt engineering or later fine-tuning as the project matures.

### Role in the Pipeline

the model receives each user query together with retrieved context passages, then synthesizes the technical guidelines into clear, non-technical, actionable guidance — without introducing unverified claims.

GROUP REFLECTION

# A Manageable, Focused Scope

The project stays narrow by focusing only on pre-flood infrastructure preparedness — not all disaster types — with a knowledge base built from a manageable set of DOST-PAGASA and OCD/NDRRMC materials, and a chatbot evaluated against a limited, well-defined test set. This keeps the project achievable within the time and resources of a student research team.